PS#11: Modeling with differential equations - Answer key

1. (AC 7.5 Exercise 7) (Skydiver).

   (a) Using k to represent the constant of proportionality, write a differential equation that describes the rate of change of the skydiver's velocity. $\frac{dv}{dt} = 9.8 - k \cdot v$.

   (b) Find any equilibrium solutions and decide whether they are stable or unstable. Your result should depend on k. Stable eq. at $v = \frac{9.8}{k}$.

   (c) Suppose that the initial velocity is zero. Find the velocity $v(t)$. $v(t) = \frac{9.8}{k}(1 - \exp(-k \cdot t))$

   (d) A typical terminal velocity for a skydiver falling face down is 54 meters per second. What is the value of k for this skydiver? $k = \frac{9.8}{54} \approx 0.1815$.

   (e) How long does it take to reach 50% of the terminal velocity? $t = \frac{\ln(2)}{0.1815} \approx 3.82$ seconds.

2. (AC 7.5 Exercise 8) (Baby weight).

   (a) Express this fact as a differential equation. $\frac{dW}{dt} = k \cdot \frac{1}{W}$.

   (b) Suppose that a baby weighs 8 pounds at birth and 9 pounds one month later. How much will he weigh at one year? $C = 64$, $k = 8.5$, $W(t) = \sqrt{17t + 64}$, and $W(12) \approx 16.37$.
   Or, if $W(1) = 10$ (more realistic), then $k = 18$, $W(t) = \sqrt{36t + 64}$, and $W(12) \approx 22.27$.

3. (AC 7.5 Exercise 9) (Salt water tank).

   (a) Since 3 gallons enter the tank every minute and 3 gallons leave every minute, what can you conclude about the volume of water in the tank?
   The volume of water in the tank is in fact constant.

   (b) How many grams of salt enter the tank every minute? 60 grams / min

   (c) Suppose that $S(t)$ denotes the number of grams of salt in the tank in minute t. How many grams are there in each gallon in minute t? $\frac{S}{100}\ \frac{\text{grams}}{\text{gallon}}$.

   (d) Since water leaves the tank at 3 gallons per minute, how many grams of salt leave the tank each minute? $\frac{3S}{100}\ \frac{\text{grams}}{\text{minute}}$.

   (e) Write a differential equation that expresses the total rate of change of S. $\frac{dS}{dt} = 60 - \frac{3S}{100}$.

   (f) Identify any equilibrium solutions and determine whether they are stable or unstable.
   $S = 2000$.

   (g) Suppose that there is initially no salt in the tank. Find the amount of salt $S(t)$ in minute t.
   $S(t) = 2000 - 2000 \cdot \exp\left(-\frac{3}{100}t\right)$